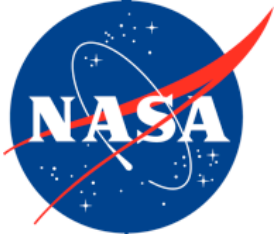


NASA/TM–20250005309



NASA's Experience Navigating the FAA's Near-Term Approval Process for UTM Services

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April 2025

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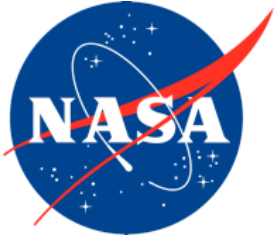
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Acronyms and Definitions

3PSP	Third Party Service Providers
ACM	Aggregate Conformance Monitoring
ACM	Aggregate Conformance Monitoring
ADS-B	Automatic Dependent Surveillance-Broadcast
AFSRB	Airworthiness and Flight Safety Review Board
AGL	above ground level
API	Application Programming Interfaces
ARC	Ames Research Center
ASTM	American Society for Testing and Materials
ATC	Air Traffic Control
ATM-X	Air Traffic Management-eXploration
AWS	Amazon Web Services
BVLOS	beyond visual line of sight
C2	Command and Control
CMSA	Conformance Monitoring and Alerting for Situational Awareness
COA	Certificate of Waiver or Authorization
ConOps	Concept of Operations
ConUse	Concept of Use
CSET	Cybersecurity Standards and Engineering Team
DAA	Detect and Avoid
DDS	Discovery and Synchronization Service
deg.....	degree
DHS	Department of Homeland Security
DoD.....	Department of Defense
DREAD.....	Damage, Reproducibility, Exploitability, Affected users, and Discoverability
DSS	Discovery and Synchronization Service
FAA	Federal Aviation Administration
FOIA	Freedom of Information Act
FPV	First Person View
ft.....	feet
GCS.....	ground control station
GPS	global positioning system
GUTMA.....	Global UTM Association
HL	Hidden Level
Hz.....	hertz (a unit of frequency equal to one cycle per second)
IETF	Internet Engineering Task Force
IMC.....	Instrument Meteorological Conditions
InterUSS	UTM/U-Space Collaboration Platform
ISO	International Organization for Standardization
KNUQ.....	Moffett Federal Airfield
KPAO	Palo Alto airport
kts.....	knots
lbs.....	pounds
LOA	Letter of Acceptance

LOB	line of bearing
MC	Mission Command
MOA	Memoranda of Agreement
MPATH	Measuring Performance for Autonomy Teaming with Humans
NAS	National Airspace System
NASA	National Aeronautics and Space Administration
NIST	National Institute of Standards and Technology
NPR	NASA Procedural Requirement
NTAP	Near-Term Approval Process
NUSS	NASA UAS Service Supplier
OEM	Original Equipment Manufacturer
OOP	Operations Over People
PCO	Pilot Computer Operator
PIC	Pilot in Command
QGC	QGroundControl
RC	remote control
RF	radio frequency
RFC	Request for Comment
RL	research laptop
RPAS	Remotely Piloted Aircraft System
RPIC	Remote Pilot in Command
RSO	Range Safety Officer
S/W	software
SAP	Software Assurance Plan
SCD	Strategic Conflict Detection
SDD	Strategic Coordination Service Description
SDMP	Software Development Management Plan
SDSP	Supplemental Data Service Providers
SEM	Service Evaluation Matrix
SLA	Service Level Agreement
SM	Smart Mobility
STRIDE	Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, and Elevation of Privilege
SUITE	Smart Urban Integrated Test Environment
SWIM	System Wide Information Management
UAS	Unmanned Aircraft System
USS	UAS Service Supplier
UTM	UAS Traffic Management
VFR	Visual Flight Rules
VLOS	visual line of sight
VMC	Visual Meteorological Conditions
VO	Visual Observer

NASA's Experience Navigating the FAA's Near-Term Approval Process for UTM Services

Abhay Borade¹, Mirna Johnson¹, Irene Smith¹, and Cynthia Wolter²

1. Introduction

The Federal Aviation Administration's (FAA) Near-Term Approval Process (NTAP) for Uncrewed Aircraft Systems (UAS) Traffic Management (UTM) third party services providers (3PSP) provides a means to evaluate and authorize the use of UTM services³. This process will expedite the approval of UTM Third Party Service Providers (3PSP) in low-risk operations to ensure National Airspace System (NAS) safety prior to rulemaking. It is not intended to be a certification process or support complex airspace operations at this time. At the successful completion of NTAP, a Letter of Acceptance (LOA) is issued to the Service Provider based on the UAS Champion Operator waiver/exemption.

The National Aeronautics and Space Administration (NASA) participated in the FAA NTAP as a UAS Service Supplier (USS), providing strategic coordination for NASA's Smart Mobility UAS operating group at NASA Ames Research Center. Smart Mobility will operate their UAS vehicles using the NASA UAS Service Supplier (NUSS) under Beyond Visual Line of Sight (BVLOS) conditions.

This technical memorandum describes NASA's experience navigating the NTAP process using the NUSS, and the NASA UAS operations team, Smart Mobility, as the Champion Operator. This document also shares feedback and responses encountered to satisfy NTAP requirements and illustrates methods of demonstrating the required criteria, which may be considered by Operators and Service Providers when pursuing an LOA through NTAP.

2. Background Information

NTAP is a pilot program that the FAA employs to determine the appropriate mechanisms for providing approval of 3PSPs. NTAP aims to approve services via exemptions sought by a Champion Operator paired with a Service Provider. An LOA will allow NASA Ames Research Center to conduct routine BVLOS operations, with the support of the NUSS, to gain situational awareness of other nearby participating operations for more efficient and safe operations. The NASA USS was the 3PSP seeking the LOA with the internal Smart Mobility organization as the Champion Operator.

¹ NASA Ames Research Center; Moffett Field, California.

² San Jose State University; Moffett Field, California.

³ https://www.faa.gov/uas/advanced_operations/traffic_management

2.1. NTAP Overview

As part of the NTAP, the FAA reviews the 3PSPs as governed by the UTM Service Provider Data Sharing and Governance Agreement. Service Providers must sign a Memoranda of Agreement (MOA) with the FAA in order to begin the review process.

To obtain the LOA, the Service Provider and the Champion Operator intending to use the service submit the following deliverables (Figure 1):

- Concept of Use (ConUse): description of the service usage and any limitations, defined and managed by the Service Provider
- Concept of Operations (ConOps): description of the operational activities, defined and managed by the Champion Operator
- Service Level Agreement (SLA): documentation of the roles and responsibilities between the Service Provider and the Champion Operator
- Signed Data Sharing and Governance Agreement
- InterUSS Automated Test Report

Additional onboarding information can be found on the GitHub for UTM Implementation (US)⁴.

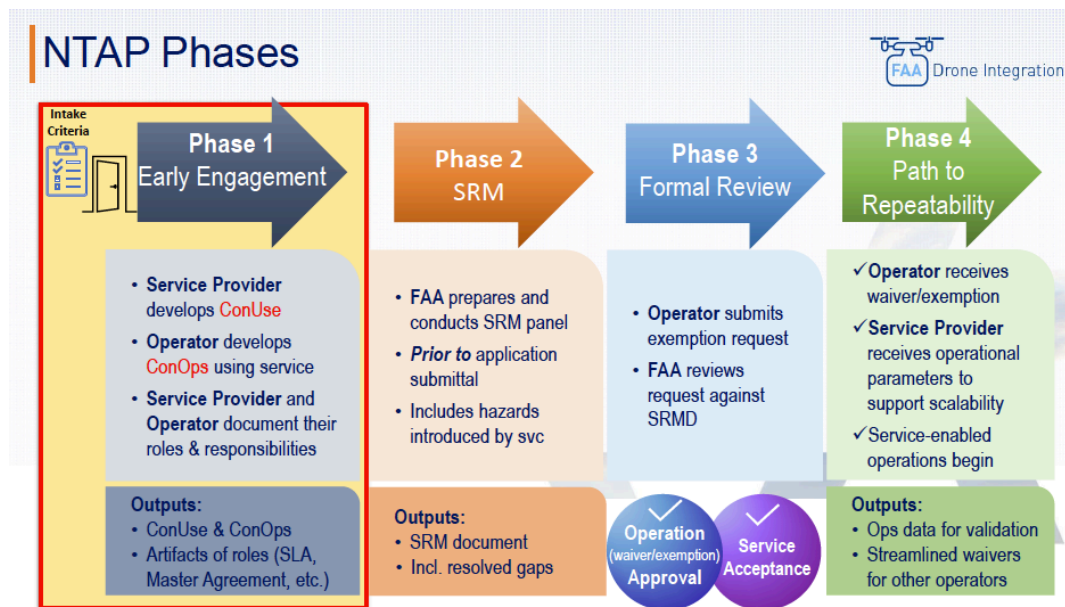


Figure 1. NTAP phases.

2.1.1. UAS Operations at NASA

UAS operations are currently being conducted by the Smart Mobility team within the NASA Ames Research Center property boundaries (Figures 2 and 3) in accordance with NASA Procedural Requirements⁵ set by the agency. NASA Ames has authorization to operate 24/7 as the received Certificate of Waiver or Authorization (COA) allows for night operations when the Operator adheres to UAS lighting requirements. Beyond Visual Line of Sight (BVLOS) operations are limited to the

⁴ <https://github.com/utmimplementationus/getstarted/blob/main/onboarding.md>

⁵ https://nodis3.gsfc.nasa.gov/displayDir.cfm?Internal_ID=N_PR_7900_003D_&page_name=main

requirements of the COA. Smart Mobility has a signed SLA with the NUSS, highlighting the operational agreements between the Service Provider and the Operator.



Figure 2. The Bay Area’s complex airspace.



Figure 3. Smart Mobility’s requested UAS operational area at Moffett Field.

2.1.2 The NASA USS

The NASS NUSS (NUSS) provided by NASA Ames Research Center was indicated as the Service Provider supporting Smart Mobility operations. This USS went through the InterUSS test suite and met the American Society for Testing Materials (ASTM) requirements⁶ agreed upon by the USS cohort as codified in the UA-UA Strategic Coordination Service Description (SDD) document⁷. Figure 4 details the ecosystem the service will be connected to as well as the connections between Smart Mobility and the NUSS. To provide context of system usage, the ConUse document included descriptions of additional elements that were part of the USS ecosystem but not necessarily the service. For example, the XtmClient was the graphic user interface serving as a bridge between the NASA operator and the NUSS. The ConUse document appears in its entirety in Appendix B and fulfilled the NASA documentation required by the FAA to meet the ConUse requirements.

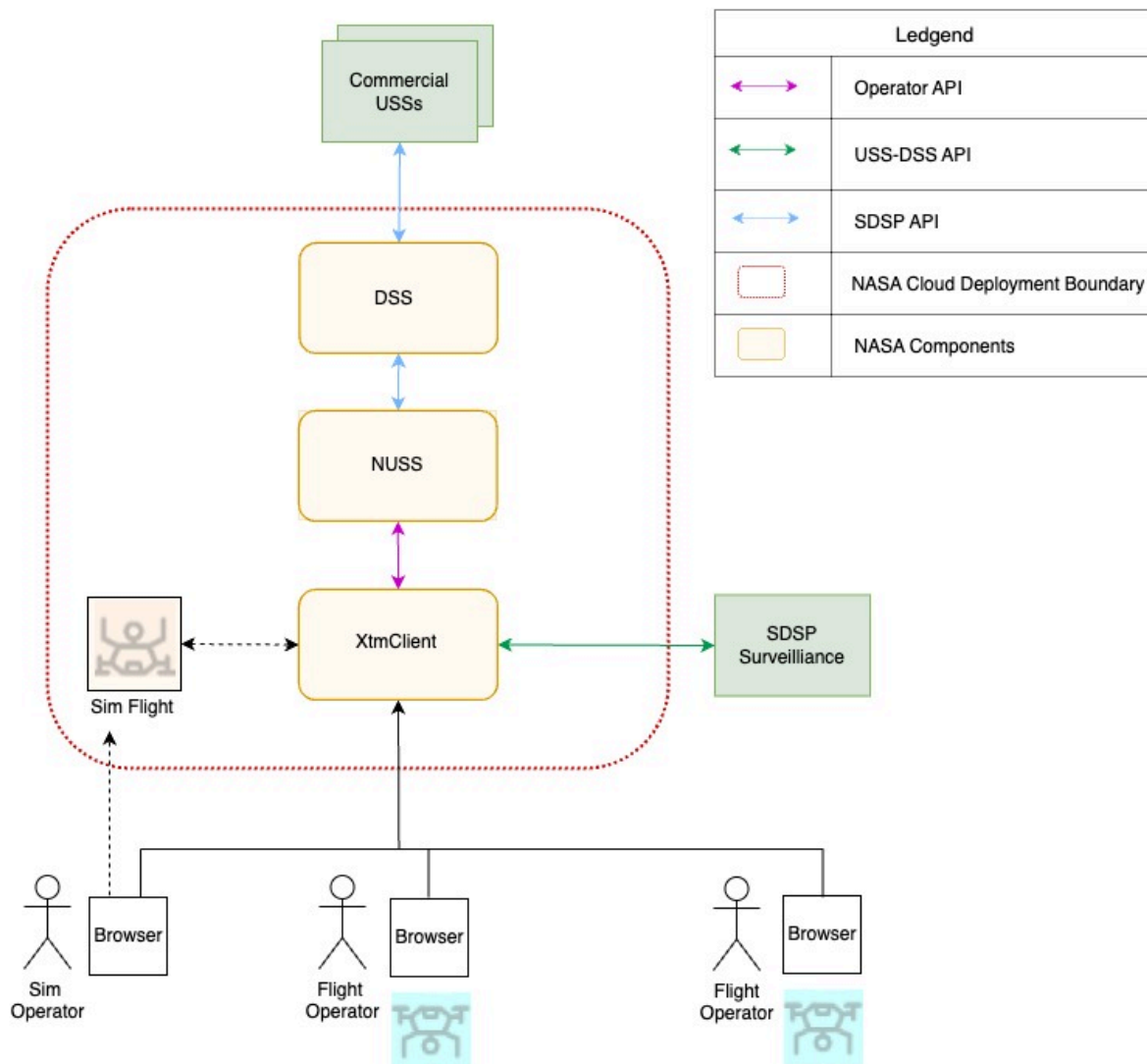


Figure 4. The NUSS operational environment.

⁶ <https://www.astm.org/f3548-21.html>

⁷ https://github.com/utmimplementationus/getstarted/blob/main/UA-UA_Strategic_Coordination_Service_Description_Document_V1.3.1.pdf/doc

3. Service Evaluation Matrix

The Service Evaluation Matrix (SEM)⁸ is used by the NTAP team to evaluate UTM services. This matrix can be used by the Service Provider to verify they are on track to meet the requirements that NTAP evaluates. It includes a description of the requirement based on the service type, rationale for the NTAP requirement, and reference to where these requirements originate, if applicable.

NASA used this scorecard to determine their readiness for NTAP and identify gaps that needed to be addressed before operating the NUSS in BVLOS operations and submitting to NTAP. The SEM suggested where compliance to the required deliverables should be recorded, and the FAA provided checklists for the ConUse, ConOps and SLA. NASA cross-checked the SEM requirements and checklists to organize the information being gathered and determine where to document compliance among the required ConUse, ConOps and SLA.

As stated in the matrix: *“The material in this document should be regarded as guidance. Service providers may propose alternative criteria, or alternative means of demonstrating a given criterion, than what is provided in this document.”*

4. Submission Artifacts

NASA submitted a NASA USS ConUse, NASA Smart Mobility ConOps for BVLOS, and the SLA to the FAA NTAP team for review. The ConUse and ConOps artifacts can be found in Appendix A and Appendix B as they were supplied to the FAA. The SLA artifact can be made available to interested readers upon request.

4.1. ConUse

The purpose of the ConUse is for the Service Provider to describe the expected usage and capabilities of the Service. It provides an interpretation for the Operator to determine whether the service will meet the needs of its users. NASA’s ConUse described the current service architecture of its USS, areas of coverage, its safety benefit and risk mitigation.

4.1.1 ISO 27001: Equivalency for Government Agencies

The ConUse documents the conformance standard the Service Provider adheres to as well as cybersecurity considerations. One of the standards required is the International Organization for Standardization (ISO) 27001 for Information Management Systems⁹, or equivalent. NASA, as a government agency, does not follow ISO 27001, but rather the widely used National Institute of Standards and Technology (NIST) SP 800-53 Security and Privacy Controls for Information Systems and Organizations. NIST SP 800-53 standardizes security and privacy controls for federal information systems and organizations to protect operations and assets, individuals, other organizations, and the Nation from a diverse set of threats and risks. All U.S. government agencies are required to follow NIST SP 800-53 and view it as an equivalent to ISO 27001. It can be expected that other federal entities also use NIST SP 800-53, and not ISO 27001. Through the NTAP, the FAA acknowledged NIST SP 800-53 equivalency to meet the information security requirements¹⁰.

⁸ https://www.faa.gov/uas/advanced_operations/USS-NTAP-Service-Evaluation-Matrix-Template

⁹ <https://www.iso.org/standard/27001>

¹⁰ <https://csrc.nist.gov/projects/olir/informative-reference-catalog/details?referenceId=99#/>

4.1.2. System Safety Assessment

NASA's initial development of their USS began with analysis of the threats to the overall UTM system using a modeling approach combination of Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, and Elevation of Privilege (STRIDE) and Damage, Reproducibility, Exploitability, Affected users, and Discoverability (DREAD) methods. National Aeronautics and Space Administration (NASA) provided a separate document for the hazard assessment outside of the ConUse¹¹. As required by the SDD, NASA followed the GUTMA Safety Task Force Safety Support Assessment¹², which includes describing contributing factors to failure conditions as a key step.

4.2. Service Level Agreement

The SLA is intended to capture the agreed-upon general responsibility between the Service Provider and the Champion Operator. It should describe how the service is intended to be implemented, and any usage limitations. This is true even when the Service Provider and Champion Operator are within the same larger organization. In this case the SLA was between the NASA USS and Smart Mobility. Due to both organizations being within NASA, the same process controls and program policies were followed. Thus, the FAA was more flexible with the NASA-to-NASA internal SLA than would be appropriate for two external organizations where more details and controls may be necessary.

4.2.1. Tailoring the Service Level Agreement

Due to NASA acting as both the Champion Operator and Service Provider, a commercial agreement was not necessary, however it was important to communicate system performance and user responsibilities through the SLA. Internally, organizations may have different processes on how work is agreed upon between different divisions, whether it be a project plan or other work agreement. For the purpose of NTAP, the SLA allows for the information to focus on technical expectations and how operations will be supported by the UTM service.

The ASTM F3623-23 Standard Specification for Surveillance Supplementary Data Service Providers¹³ (SDSPs) has an outline for an SLA that can be leveraged. NASA worked with the FAA to tailor an SLA outline that was better applicable for a USS. One of the main differences for an SLA with a USS, is that rather than describing data performance metrics that are more applicable to an SDSP, two sections were added to address Roles and Responsibilities as well as Safety Protocols and Response Plans. NASA further recommends keeping a performance-related section in the SLA to capture metrics available to monitor and/or test system performance of the USS.

The SLA may need to describe information that is already in the Service Provider's ConUse. Since the ConUse is also part of the NTAP submission process, referencing sections from the ConUse in the SLA will also make the SLA easier to document and avoid duplication or inconsistencies.

4.2.2. SLA Ownership

As NASA looked at both current internal UTM operations and future operations within the agency, the need to reuse the SLA became evident. As a Service Provider that expects to offer the same service to multiple Operators, creating an operator-agnostic SLA will be more efficient in communicating consistent responsibilities. NASA's SLA reflects an operator-agnostic SLA that standardizes shared responsibilities, expectations from the Operator, and the Service Provider.

¹¹ <https://arc.aiaa.org/doi/epdf/10.2514/6.2017-3272>

¹² <https://gutma.org/blog/2024/03/18/gutma-safety-task-force-releases-ussps-safety-support-assessment/>

¹³ <https://www.astm.org/f3623-23.html>

4.2.3. Service Provider Responsibilities for the USS

In both the ConOps and ConUse, geographic region is described. In the ConUse, NASA describes the region the service is capable of supporting. In the ConOps, the region the Champion Operator intends to operate is described. Expectedly, there is an overlap between the two but the responsibility of ensuring that the operations using the service remain in that overlapping region has been unclear in the UTM community. Since the Operator obtains the COA, it is in the best position to know where it can and cannot operate. Similarly, in reviewing the ConUse of the service, the Operator is in the best position to determine if the service's system performance is adequate to support its operations in a given region. Both the status of the COA and operational goals can be dynamic that the USS is not purview to, thus, NASA's SLA outlined that ensuring geographic conformance of operations while using the service as an Operator responsibility.

4.3. ConOps

The purpose of the ConOps is to describe the operations and the intended use of systems to support operational needs. This should include justification for the use of the UTM service seeking an LOA for BVLOS, as well as a description of the UAS specifications, even for Commercial Off-the-Shelf vehicles. The ConOps is created and managed by the Champion Operator. Below are subsections of the ConOps where NASA obtained further clarification or direction from the FAA.

4.3.1. Roles and Responsibilities

It is important that all parties, including Service Providers and the FAA NTAP committee, understand the crew personnel involved, described by role type. Specifically, this should include the responsibilities that each role has and the training and certifications that are required. NASA Smart Mobility ConOps for BVLOS provides an example of how this was described.

4.3.2. Command and Control

The ConOps should describe all tools to perform command and control during operations, and not only the Command and Control (C2) link.

4.3.3. Emergency Procedures

Since NASA was using commercially available UAS, NASA followed the manufacturer's recommended procedures for handling aircraft in the event of an aircraft emergency. The manufacturer's documents and website were referenced and deemed sufficient. Note that a manufacturer may update their procedures and the Operator should be vigilant in staying up to date on the manufacturer's latest versions.

4.3.4. Human Impact Testing

The FAA uses this type of analysis and risk assessment to determine if a Safety Risk Management review of the service is required. While UAS manufacturers publish significant information about their products, some information may not be readily available, as was encountered for the Human Impact Testing that the vehicles Smart Mobility at NASA Ames is using. In that case, NASA was able to reference a Risk Assessment for Small UAS¹⁴, identifying the relevant UAV class and the operation configuration to determine the severity and safety objectives.

¹⁴ <https://arc.aiaa.org/doi/epdf/10.2514/6.2017-3272>

4.3.5. InterUSS Test Report

Upon signing the Governance Agreement, applicants need to complete the official automated testing. The SDD defines the baseline identifier for the test configuration that Service Providers should be using. Upon initial review of the UTM/U-Space Collaboration Platform (InterUSS) test report, the FAA immediately looks for the baseline version used to test to ensure they match.

5. Governance Agreement

Signing a Data Sharing and Governance agreement (based on the SDD) is requires Service Providers, specifically in the United States, enter a Data Sharing and Governance Agreement based on the SDD. This cooperative agreement is between Service Providers and Operators and includes how they share data, govern operations in shared airspace, and manage conflicts between UAS. Signing this agreement is required for FAA's NTAP before the Service Providers can be eligible for an LOA and obtain access to a Remotely Piloted Aircraft System (RPAS) platform to conduct automated testing for UTM interoperability.

5.1. Considerations for Government Agencies

NASA is the first federal agency to review the Data Sharing and Governance Agreement with the intent to sign and operate in shared airspace. At the time of NASA's NTAP application, there were several portions of the agreement that NASA, and likely other federal agencies, would not be able to agree to and prevent signing the agreement. For example, clauses related to confidentiality would be in conflict with U.S. Government laws as the Freedom of Information Act (FOIA)¹⁵ allows the public to request records from the government. Additionally, requiring indemnification, liabilities, warranties and representation do not fit the context for a federal agency. At the time of this writing, NASA is working on a proposal to address this issue such that federal agencies can enter the agreement.

6. Changes after an Initial LOA

As NASA considered long term concepts for expanding UTM BVLOS, LOA change management scenarios were identified and discussed. These included adding a new Operator after issuance of an LOA and an Operator championing an additional service (Figures 5–7).

¹⁵ <https://www.foia.gov/foia-statute.html>

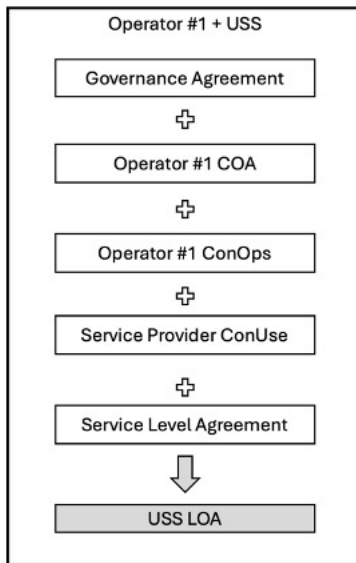


Figure 5. Original LOA process.

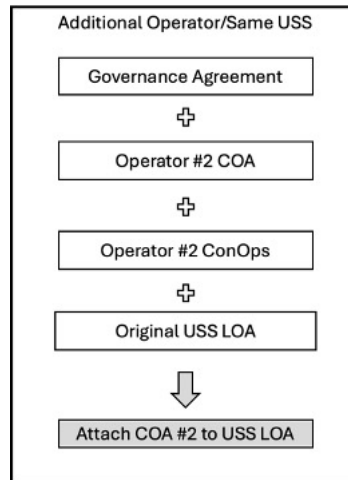


Figure 6. Additional Operator after LOA.

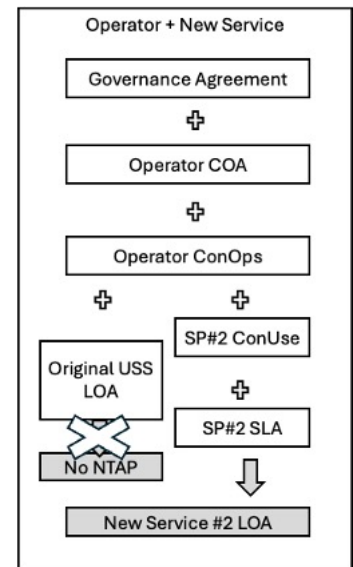


Figure 7. Operator adds a new service.

Appendix A. NASA for Smart Mobility UTM BVLOS Operations

Concept of Operations

UTM BVLOS Sub-Project

August 2024

(This outline was generated using the ConUse ConOps SLA Score Template provided by the FAA for the Near-Term Approval Process.)

1. Introduction

Smart Mobility (SM) at Ames is an initiative within the NASA Ames Research Center Aeronautics Project Office (Code AT) created to enable aeronautics research activities that involve live flight of vehicles. SM was created based on a need for unmanned aircraft system (UAS) subject matter expertise on aeronautics projects wishing to fly at Ames.

The SM architecture uses the NASA USS as a mean to capture and preserve real-time sUAS telemetry data as well as contextual messages, operations and surveillance data generated during flight activities. Data sources include UAS sensors (e.g. navigation, vehicle status, telemetry) and ground instrumentation (e.g. radar, RF, weather, ADS-B antennas, etc.); additional data sources as required by the mission.

Smart Mobility will be the champion operator that will be using the NASA USS. By doing so they will gain situational awareness tools in order to efficiently and safely operate BVLOS at Ames. These tools include but not limited to Strategic Conflict Detection, Conformance Monitoring and messaging as per NASA USS ConUse. This adds a layer of safety to the outside world and other USS nearby for a higher level of situational awareness.

1.1. Mission, Scope and Goals

To conduct BVLOS operations at Ames, which is situated inside the Moffett Federal Airfield (KNUQ) Class D airspace by leveraging the NASA USS.

1.2. Requested Waivers, Authorizations or COAs

sUAS operations at NASA ARC are currently done mainly utilizing the 2020-P107-WSA-18993-R Airspace Authorization for Part 107 operations.

Requesting waivers for 14 CFR 107.31 (visual line of sight aircraft operation) and 14 CFR 107.33 (visual observer).

2. Unmanned Aircraft Specifications

3D-view of drawings or photographs of the Unmanned Aircraft (UA).

The SM team currently operates two vehicles fully integrated into the Build 1 architecture: the Alta 8 Pro and the Alta X models, which are commercial off-the-shelf UAS manufactured by Freefly Systems. The Freefly Alta UAS use the Alta-specific version of the QGroundControl (QGC) software for the ground control station (GCS). Through its connection to the UAS, the Research Laptop acquires sUAS telemetry and global positioning system (GPS) flight data in real time.

2.1. Unmanned Aircraft Specifications.

The Smart Mobility at Ames team (SM) currently operates the following vehicles.

2.1.1. Freely Alta X



Freely Alta X Specifications	
Manufacturer	Freely Systems
Country of Origin	U.S.A.
UA Geometry	90 in with unfolded arms and with props
Gross Weight (lbs)	Without payload: 44.1 lbs With maximum payload: 76.9 lbs (Part 107 operations max weight <55 lbs observed)
Payload	Max: 35lb
Payload Type	Various Camera or custom research payload
Aircraft Type	Quadcopter
Thrust System	Electric
Motor or Engine Spec	Motor Max Continuous Power: 17,760W Motor Max Instantaneous Peak Power: 23,088W (<3s) Kv: 115 rpm/V DJI
Propeller Spec	Size: 33" x 9"
Min/Max Operating Alt (AGL)	Not stated by manufacturer, but at least 10,000 ft pressure altitude (payload dependent)
Max Cruise Speed (knots)	39 kts
Fuel	Electric
Battery System	2 x 12S1P 16 Ah LiPo batteries (no backup)
Unique Features	FPV Camera; ADS-B In added feature
Endurance	Max hover time, 0lbs, Payload: 53 min
Range in Statute Miles	39kts * 53 min = 34NM = 29 SM
Remote Identification	Yes

2.1.2. Freely Alta 8 Pro



Freely Alta 8 Pro Specifications	
Country of Origin	U.S.A.
UA Geometry	68 inches with unfolded arms and with props
Gross Weight (lbs)	Without payload: 22.4 lbs (with heaviest batteries) Without payload: 19.6 lbs (with lightest batteries) With maximum payload: 40.0 lbs.
Payload	Max: 20lb
Payload Type	Various Camera or custom research payload
Aircraft Type	Octocopter
Thrust System	Electric
Motor or Engine Spec	Motor Max Continuous Power Output: 350 W Motor Max Instantaneous Peak Power Output: 950 W Kv: 384 rpm/V Freely
Propeller Spec	Size: 18" x 6"
Min/Max Operating Alt (AGL)	Maximum service ceiling: Not stated by manufacturer, but at least 10,000 ft pressure altitude (payload dependent)
Max Cruise Speed (knots)	39 kts
Fuel	Electric
Battery System	2 x 6S1P 16 Ah LiPo batteries (no backup)
Unique Features	FPV Camera
Endurance	Max hover time, 0lbs, Payload: 34 min
Range in Statute Miles	39kts * 34 min = 22NM = 25 SM
Remote Identification	Yes

2.1.3. Skydio X2



Skydio X2 Specifications	
Manufacturer	Skydio Inc
Country of Origin	U.S.A.
UA Geometry	26.1 inches by 22.4 inches with unfolded arms and with props
Gross Weight (lbs)	Without payload: 2.92 lbs. With maximum payload: 3.37 lbs
Payload	Max payload: 0.44 lbs.
Payload Type	Various custom research payloads
Human impact testing data (aircraft and/or payload)	Data not found
Aircraft Type	Quadcopter
Thrust System	Electric
Motor or Engine Spec	Not Specified
Propeller Spec	Not Specified
Min/Max Operating Alt (AGL)	Maximum service ceiling: 12,000 ft.
Max Cruise Speed (knots)	22 kts
Fuel	Electric
Battery System	Skydio (11.55 VDC, 8200 mAh) LiPo batteries (no back up)
Unique Features	Obstacle avoidance: Stationary objects, omnidirectional visual. Lights: Strobes, visible at 3 SM
Endurance	35 min
Range in Statute Miles	22 kts * 35 min = 13 NM = 15 SM
Remote Identification	Yes

2.1.4. Skydio X10



Skydio X10 Specifications	
Manufacturer	Skydio Inc
Country of Origin	U.S.A.
UA Geometry	31.1 x 25.6 inches with unfolded arms and with props
Gross Weight (lbs)	Without payload: 4.72 lbs. With maximum payload: 5.49 lbs
Payload	Max payload: 0.77 lbs
Payload Type	Spotlight Speaker and Mic Parachute RTK/PPK GPS NightSense Visible/IR light. Various custom research payloads
Aircraft Type	Quadcopter
Thrust System	Electric
Motor or Engine Spec	Not Specified
Propeller Spec	Not Specified
Min/Max Operating Alt (AGL)	Maximum service ceiling: 15,000 ft.
Max Cruise Speed (knots)	39 kts
Fuel	Electric
Battery System	Skydio - One (18.55 VDC, 8419 mAh) LiPo batteries (no backup)
Unique Features	Obstacle avoidance: Stationary objects, omnidirectional visual Optional accessory/payload: - Spotlight - Speaker and Mic - Parachute - RTK/PPK GPS - NightSense Visible/IR light.
Endurance	40 min
Range in Statute Miles	39 kts * 40 min = 26 NM = 30 SM
Remote Identification	Yes

3. Flight Limitations

3.1. Operating Rules and Meteorological Conditions

<i>Operating Rules and Conditions</i>	<i>Operation Sought</i>
Visual Flight Rules (VFR)	Yes, as directed by nearby ATC-provided weather data (METAR) and local on-site weather assessments.
Visual Meteorological Conditions (VMC)	Assessed from nearby ATC-provided weather data.
Instrument Flight Rules (IFR)	No
Instrument Meteorological Conditions (IMC)	Assessed from nearby ATC-provided weather data.
Flight through visible moisture (rain, snow, clouds, fog, mist etc.)	No
Flight operations in known icing conditions	No
Wind limitations (max crosswind, takeoff and landing, enroute cruise etc.)	12G18

3.2. Types of Operations

<i>Type of Operation</i>	<i>Operation Sought</i>
Visual Line of Sight (VLOS)	Currently in place (2020-P107-WSA-18993-R)
Beyond Visual Line of Sight (BVLOS)	Yes
Operations over People (OOP)	Yes
Daylight	Yes
Nighttime	No
Operations from a moving vehicle or vessel	No

4. Crew and Personnel

4.1. Minimum Crew

- 1x Remote Pilot in Command (RPIC)
 - 1x Pilot-Remote in Command (PRCO) by Skydio
- 1x Pilot Computer Operator (PCO) at the Ground Control Station
- 1x Range Safety Officer (RSO)/Visual Observer (VO) if required by NASA Procedural Requirements (NPR) 7900 or by the COA

4.2. Roles and Responsibilities

4.2.1. Range Safety Officer

In accordance with NPR7900.3D. “5.1.4.7 Range Safety Officer (RSO). The RSO provides duties as assigned in accordance with NPR 8715.5 levied on UAS operations during FRR/Airworthiness and Flight Safety Review Board (AFSRB) reviews and during operations as required.”

4.2.2. Observers

Any observers are added to the flight crew to increase the situational awareness of the flight crew in the operational areas, and the need depends on the local lay of the land and the activities going on around the area.

4.2.3. Mission Commander

In accordance with NPR7900.3D. “5.1.4.5 UAS Mission Commander (MC). The UAS MC exercises command over single NASA UAS or a formation of UAS. He/she is responsible for all phases of the assigned mission except those aspects of safety of flight that are related to the physical control of the aircraft and fall within the prerogatives of the PIC (unless the PIC and MC are the same individual).”

4.2.4. Training

RPICs maintains a current FAA P107 certification and NASA ARC established training curriculum, including annual curriculum and checkride currency, as well as a 90-day flight operations currency throughout the year.

4.2.5. Communication

During flight operations the flight operators (PRCOs and PCOs) are in contact with each other and any Mission Commander (MC) and Range Safety Officer (RSO) through face-to-face communication, or by using cell phone and/or VHF radio links, when needed due to locational separation. The RSO, or other designated range safety responsible person, is also in communication with local ATC, in accordance with established COA.

4.2.6. Physical Security and Control

Operations are conducted inside ARC boundary fence, which has access control by base security checkpoints. Within the ARC boundary fence, the designated flight operations area will have road closures and warning signs at access points, for center personnel awareness.

4.2.7. Hand-Off Procedures

For handoff between remote control (RC) pilot and ground control station (GCS) operator, a three-way handoff is utilized, as is done in a crewed aircraft cockpit between two pilots.

4.2.8. Safety of Personnel

- The designated flight operations area is cleared of non-participating persons.
- A general 45-deg rule in place, where the aircrafts altitude in AGL represents a safety buffer stand back range, such that anyone viewing the aircraft at a higher elevation than 45 deg are too close and need to stand back, or under a protective covering.
- Utilization of personal protective equipment (PPE) (e.g. safety glasses, gloves, covering clothing and shoes) for those approaching a powered aircraft.

4.2.9. Training of Crew

All personnel present at the flight operations, including non-flight crew, take part in the pre-flight briefing where objectives and safety aspects of the day is briefed, and specific roles are delegated and explained. Part of this is utilizing all persons involved as observers, and in how to report any perceived issues or concerns during flight operations.

<i>Crew</i>	<i>Training</i>
Operator (PRCO and PCO)	In accordance with established ARC UAS operations training requirements with initial and annual recurrency (unless stated otherwise): • Crew Resource Management training, no less than 4 hrs. • Ground school reviewing applicable NASA Procedural Requirements (NPR), Ames Procedural Requirements (APR), SATERN training videos, and related FAA COAs, with written exam to pass. • Part 107 certification and biannual currency in accordance with FAA regulations. • Successfully completion of checkride according to specific maneuvers. • 90-day 3-TOL for currency.
RSO	Follow Training requirements per NPR 7900. In accordance with training plan laid out in QS-TR-2240B Range Safety Personnel Training Plan 2021 10-14 document.
VO	Follow training requirements per NPR 7900 and Part 107 training requirements.

5. Ground Station and Support Equipment

Smart Mobility team has developed an integrated test infrastructure, named the Smart Urban Integrated Test Environment (SUITE), currently in its Build 1 version. The sUAS is connected via a GCS that communicates to a research laptop (RL). The RL is connected to the NASA USS environment through a web interface called XTM Client, which provides the required USS services for the task. In addition to sUAS telemetry/GPS flight data, Build 1 integrates a passive surveillance tracking capability, provided by Hidden Level, Inc. (HL), as the USS Supplemental Data Service Supplier (SDSP), that detects not only the subject UAS but also surrounding air traffic. Additional sensors include passive RF-signal sensors, an LSTAR[®] radar, an ADS-B Ping Station and not-networked wind sensors along the flight corridor.

5.1. Location of Ground Control Station

Describe the location of the ground control station (enclosed vehicle, open air, etc.).

- The Skydio UAS have the GCS built into the RC unit and thus is managed by the PRCO as well.
- When applicable (Alta X and Alta 8 Pro), a separate Ground Control Station is predominantly situated in an enclosure (vehicle, trailer, building) or infrequently in open air if operational circumstances require.



Figure A1. Interior of SM trailer with ground crew workspace.



Figure A2. Workstations of Flight Data Engineer operating the research laptop (left) and PCO operating the GCS (right).

5.2. Configuration of Ground Control Station

Alta X and Alta 8 Pro utilize a laptop computer and Ground Control software with a 900 MHz ISM radio link to the aircraft. To connect to the NASA USS and log flight operations data, the flight data engineer uses a separate Research Laptop logged into a secure web interface called XTM Client, which will record information related to messages, operations, and position / surveillance data.

Skydio GCS is built into the RC unit, and NASA has future plans to connect this to NASA USS.

5.2.1. Hidden Level

The Hidden Level sensor network consists of a distributed set of HL1000 passive RF sensors. Each HL1000 sensor independently scans a broad range of frequencies, searching for UAS based on their command and control, video, or telemetry links. Once a UAS has been detected the sensor determines a line of bearing (LOB) in both azimuth and elevation from the sensor to the UAS and shares the LOB information, along with any other pertinent identifying information, with the AMS for further processing. From an AMS data consumer's point of view, these sensors are transparent to them as Hidden Level handles the data interconnect between systems and the AMS. Smart Mobility operates while using Hidden Level provided data of the surrounding airspace for detection of other

UAs nearby, and if detected, mitigate any risks of in-air collisions by taking appropriate evasive maneuvers or stand down operations until potential conflicts have cleared.

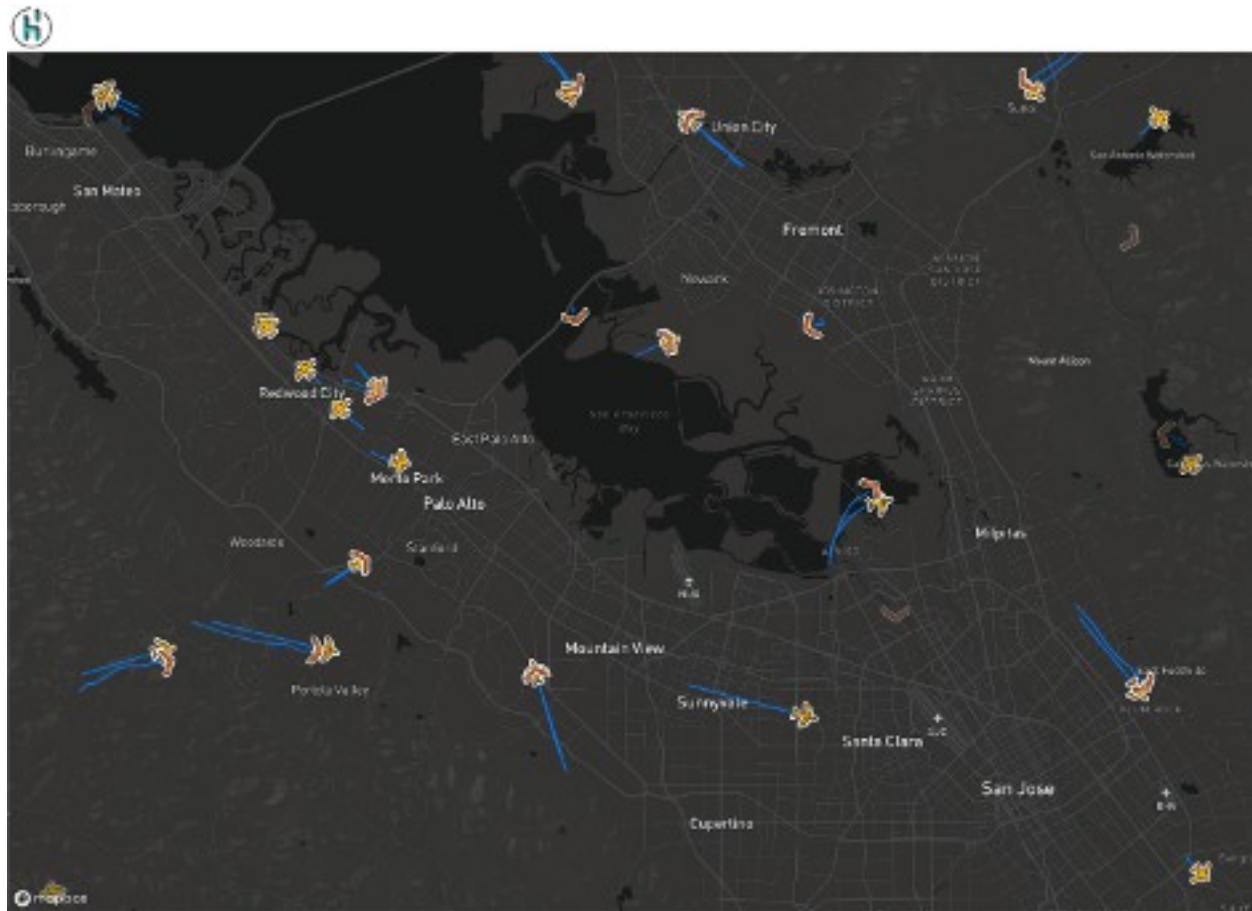


Figure A3. UAS detected by Hidden Level sensors.

5.2.2. Support Equipment

- Takeoff and landing pad
- Airband VHF radio
- Handheld wind meter
- PPE (safety glasses, gloves etc.)
- Safety vest and warning signs

6. Command and Control

6.1. Command and Communication Functions

Describe plans involving command and communication functions between different portions of the UAS.

- a. Unmanned aircraft
 - All UAS have a RC function through which an operator (PRCO) can take over directly and manually guide the UA through the RC stick inputs, if needed.
- b. Control station
 - The Skydio X2 and X10 UAS have the RC and GCS elements combined into one handheld unit. The RC operator (PRCO) is therefore also conducting the functions of a GCS operator (PCO).
 - The Freefly Alta X and 8 Pro configurations can be set up in a few optional ways:
 - 1x RC unit controlled by one RPIC (NASA term PRCO).
 - 1x RC unit (PRCO) and 1x GCS (PCO) unit operated by separate pilots, and PIC role can be transferred in-between through a three-way handoff communication.
 - 1x combined RC and GCS unit operated by one RPIC.

6.2. C2 Command Control Link

Define the C2 command and control link. This includes but not limited to:

- a. The electronic security of the control link.

<i>UAS</i>	<i>Electronic Security of Control Link</i>
Freefly Alta	None other than paired radio units
Skydio X2	AES-256
Skydio X10	AES-256 (5G: AES-128)

- b. The real time situational awareness such as RangeVue, DAA, FPV, SWIM, etc.
 - All UAS:
 - NASA USS implementation with sensors covering the designated flight operations area:
 - * Ground-based ADS-B In
 - * Hidden Level passive sensors
 - * L-STAR active radar
 - Freefly Alta X: Onboard ADS-B In, displaying traffic on the GCS.
 - Other UAS: no additional
- c. Loss of positive control procedures such as lost link, fly away, and C2 link latency and reliability.

<i>UAS</i>	<i>Procedure for Loss of Positive Control</i>
Freefly Alta X and 8 Pro	With Failsafe (option of Return to Home, Land, Hold) • C lost link: after 0.5 s (variable) • GCS lost link: 10 s (variable)
Skydio	RC lost link: Return to Home

- d. Emergency procedures (return to base, hover, grounding, infrastructure masking/shadowing).
 - RPIC commands a Return to Home or Land, as appropriate for the emergency at hand
- e. Uplink and downlink information (operating frequency and range, backup, etc.).

<i>UAS</i>	<i>Uplink and Downlink Information</i>
Freefly Alta X and 8 Pro	2.4 GHz (RC), range not specified. 900 MHz (GCS), range >5000 ft
Skydio X2	5.8 GHz, range > 3.7 miles
Skydio X10	Connect SL: 7.5 miles. 2400-2483.5MHz, 5150-5850Mhz Cellular 5G: Unlimited (wherever cellular coverage is available) 600MHz-4400MHz

6.3. Additional Communication Links

Any additional communication links (video, payload, etc.).

- Freefly Alta X and 8 Pro: 5.8 GHz (FPV), unspecified range.

6.4. Software Information

<i>Application</i>	<i>Software Information</i>
Freefly Alta X and 8 Pro	OEM software Optional for GCS: MPATH, NASA Langley customized version to QGC s/w provided by OEM
Skydio X2 and X10	OEM software
NASA USS	NASA-developed software Reference NASA USS ConUse
Telemetry Module	NASA-developed Software Reference NASA USS ConUse No effect on Command and Control if it fails

7. Execution of Operations

7.1. Mission Limitations

The flight missions will be contained within the designated flight operations area and will vary in nature depending on various projects and payloads.

7.2. Geography

The proposed BVLOS designated flight operations area is situated within the current COA borders and KNUQ Class D airspace. Flight operations will be in coordination with local ATC. Low Altitude Authorization and Notification Capability (LAANC) designation in a “0” grid space, however, 2020-P107-WSA-18993-R Airspace Authorization allows for sUAS operations up to 400 ft (within light-blue area in Figure 8). Palo Alto (KPAO) occasionally extends their traffic pattern in the KNUQ. Coordination will take place with KNUQ.

See boundaries of the proposed BVLOS designated flight operations area outlined in blue.



Figure A4. Map over NASA Ames and KNUQ depicting boundaries of the P107 COA (light blue, long term boundary) (- 122.037606,37.401814, -122.034672,37.426292, -122.01705,37.445411, -122.068386,37.434906, -122.068861,37.408525, -122.037606,37.401814).

7.3. Launch, Fly, Recover

Describe launch/fly/recover aspects of the operations.

a. Give clear and concise explanation of launch and recovery.

- How will launch and recovery be accomplished?

- Launch

- * The UAS checklist is completed, which includes local notifications and system checks.
- * Designated RSO approval for flight.
- * The UA is set up on the orange launch pad and the area around the UA is cleared of any personnel (to at least the 45-deg range of the initial takeoff altitude).
- * “Aircraft launching” is loudly announced by the PIC.
- * Vehicle is armed and if everything seems normal, launched vertically to an initial launch altitude, e.g., 10 ft.
- * A basic functional check of the UAS is performed, to verify correct and accurate control.
- * If satisfied, the mission will be engaged.

- Recovery

- * When mission is completed, and the vehicle is returning the PIC will loudly announce “Aircraft returning.”
- * The recovery site is cleared of any personnel out to at least the 45-deg range.
- * The local wind conditions will be assessed for timing and any needed adjustment of landing heading.

- * UA will make a descent over the recovery site and any necessary adjustment to its position will be made by the PIC.
- * The PIC will ensure a smooth and precise touch down and then disarm the UA.
- * PIC will announce “Clear to approach” when the UA has been secured and is safe to approach by others.
- *Where will launch and recover be accomplished?*
 - Launch and recovery will be accomplished at specified and marked sites within the designated operations area, as briefed to flight operations participants in the pre-flight briefing.
- b. *Will your UA launch/fly/recover only over private property with owner’s permission?*
 - No, only over NASA Ames property.
- c. *Identify the congestion (demographics, population per square mile, urban, rural etc.).*
 - The designated flight operations area is over an urban environment (buildings, cars, office space). However, operations are being conducted in less densely populated areas within NASA Ames property.
- d. *Identify and describe the UA’s proximity to airports, agricultural, heliports, airports, electrical grid etc.*
 - The UA will be operating within the KNUQ Class D airspace and the BVLOS designated flight operations area is adjacent to the KNUQ airfield.
- e. *Identify whether flights are planned over people not involved in the operation.*
 - Flights are planned over people not involved in the operation.
 - UAS operational intents are not intentionally planned over people, however mitigations in place to reduce risk in case overlaps occur.
 - To minimize intentionally flying over people, measures are in place such as flying over less populated areas, close roads, fly over water, use geofencing to prevent fly over etc. No loitering over events but will transition over people.
- f. *Describe the UA’s proximity to people, infrastructure, roads, vehicles and/or vessels.*
 - The UA will be operating over roads within the NASA Ames property that will be closed off for access during the flights.
 - There are NASA Ames owned buildings and infrastructure within the proposed BVLOS designated flight operations area.
 - There is a publicly accessible path outside the designated flight operations area and the NASA Ames boundary fence to the west.
- g. *How are current weather conditions determined at departure, arrival, and enroute?*
 - The local weather conditions, and reports from KNUQ, are monitored before launch, during flight, and before return of the UA.

7.4. Auto-Pilot Level

Operations conducted with 1 operator to 1 UAS (1:1), or 2:1 where a separate RC operator (PRCO) and GCS operator (PCO) are used. No request for exclusive airspace.

7.5. Coordination and Communications Required

- a. Describe coordination and communications required to conduct the operations.*
 - Proposed flight operation review and approval by applicable NASA Ames review boards:
 - Airworthiness and Flight Safety Review Board, on UAS airworthiness.
 - Flight Readiness Review Board, on the proposed flight operations.
 - Mission Readiness Review Board, on multi-vehicle operations.
 - Pre-ops concurrence from NASA Ames Flight Operations Management Division (Chief of Flight Operations approval).
 - Designated flight operations area stakeholder notifications, including KNUQ ATC.
 - Pre-flight briefing for flight crew and any personnel involved in the operations.
 - Designated RSO pre-launch approval.
- b. Describe communications expectations with ATC if applicable.*
 - In accordance with current COA, 3-day preflight notification as well as notification day of through cell before flight ops and again after conclusion of flight ops.
- c. Describe community outreach plans (flying / non-flying public, municipalities, airports, etc.).*
 - Nonspecific, ops occurring on NASA Ames property.
- d. Describe if/when flight plans will be filed with ATC (VFR/IFR).*
 - Designated flight operations area is included in the ATC communications mentioned above, as well as maximum operations altitude.
- e. Identify liaisons with ATC (if not required, clearly state so).*
 - In accordance with current COA (see above).
- f. Identify mishap reporting procedures such as an accident/incident response plan and/or FAA/NTSB notification*
 - As specified in the Safety and Mishap Preparedness and Contingency Plan, part of the Flight Readiness Review Board review documentation.
 - This defines the plan for responding to and reporting mishaps in accordance with NPR 8621.1D NASA Procedural Requirements for Mishap and Close Call Reporting, Investigating, and Recordkeeping, and Ames Health and Safety Manual, Chapter 4, Mishap and Close Call Reporting and Investigating/Contingency Plan.
- g. Identify when any Notices to Airman (NOTAM) will be posted.*
 - When directed through COA or by ATC.

Appendix B. NASA UAS Service Supplier (USS)

Concept of Operations

UTM BVLOS Sub-Project

September 2024

General Information

NASA USS is a UAS Service Provider software system designed and built by Aviation Systems Division at NASA Ames Research Center. NASA USS provides a platform to enable and advance the research collaboration on UTM (Unmanned Aircraft System Traffic Management) with the industry, government agency, and academia.

NASA UTM services are based on digital sharing of each user's planned flight details, providing the same situational awareness of the airspace to each user. It aims to accommodate a variety of business models vehicle types and airspace uses, such as package delivery, inspections, public safety, and security at NASA facilities.

NASA USS is an implementation of UTM services as per ASTM F3548 standard. NASA USS is based on past work from NASA UTM Project Technical Capability Level 1–4 demonstrations, NASA UAM Airspace X4 simulation and various other NASA Air Traffic Management-eXploration (ATM-X) research. These projects included partners from the Federal Aviation Administration, industry and academia, and that tested technologies for an eventual traffic management system.

NASA USS is currently being developed by the Federated Airspace Management Framework (Freddie) team. NASA Ames Research Center established Freddie as an implementation of an advanced, collaborative, digital, and automated approach to managing air traffic within the airspace. Freddie has been used internally and externally at NASA to evaluate UTM work. NASA USS has taken part in various field tests involving the industry, academia, Department of Defense (DoD), Department of Homeland Security (DHS), FAA, and others.

Freddie deploys its major platform functionalities as micro-services on AWS cloud services for scalability and customizability. The hardware associated operating cost can be managed effectively through AWS by scaling up or down according to the demand.

1. Summary of UTM Services

NASA USS provides several services to manage drone traffic efficiently and safely in low-altitude airspace. The main service provided to the operator is Strategic Coordination to the Operator per ASTM- F3548-21.

1.1. Architecture USS and Discovery and Synchronization Service

To enable federated traffic management via a collection of automation-enabled entities, those entities require a way to discover how to connect with each other. NASA tested approaches to this discovery process during previous flight testing and then industry developed improved approaches, including one called Discovery and Synchronization Service (DSS). DSS is currently being developed to meet the requirements of emerging standards. DSS was tested in various forms during NASA flight testing FAA flight testing and other international flight test events.

The DSS¹⁶, hosted by InterUSS Platform, allows USSs to communicate information about UAS operations and coordinate with each other. It works like a phonebook which allows a USS which is flying over a particular geographic area, to find other USSs in the same area. As per the specification, those USSs participate in a dialog to notify others USS in the same geographic area, by sharing their operational intents. NASA will deploy its own DSS for NASA USS.

Only NASA UAS operators can connect to use the NASA USS for their operations. A NASA UAS team will connect their vehicle to a GCS and have the USS information available for the crew to use for deconfliction and coordination.

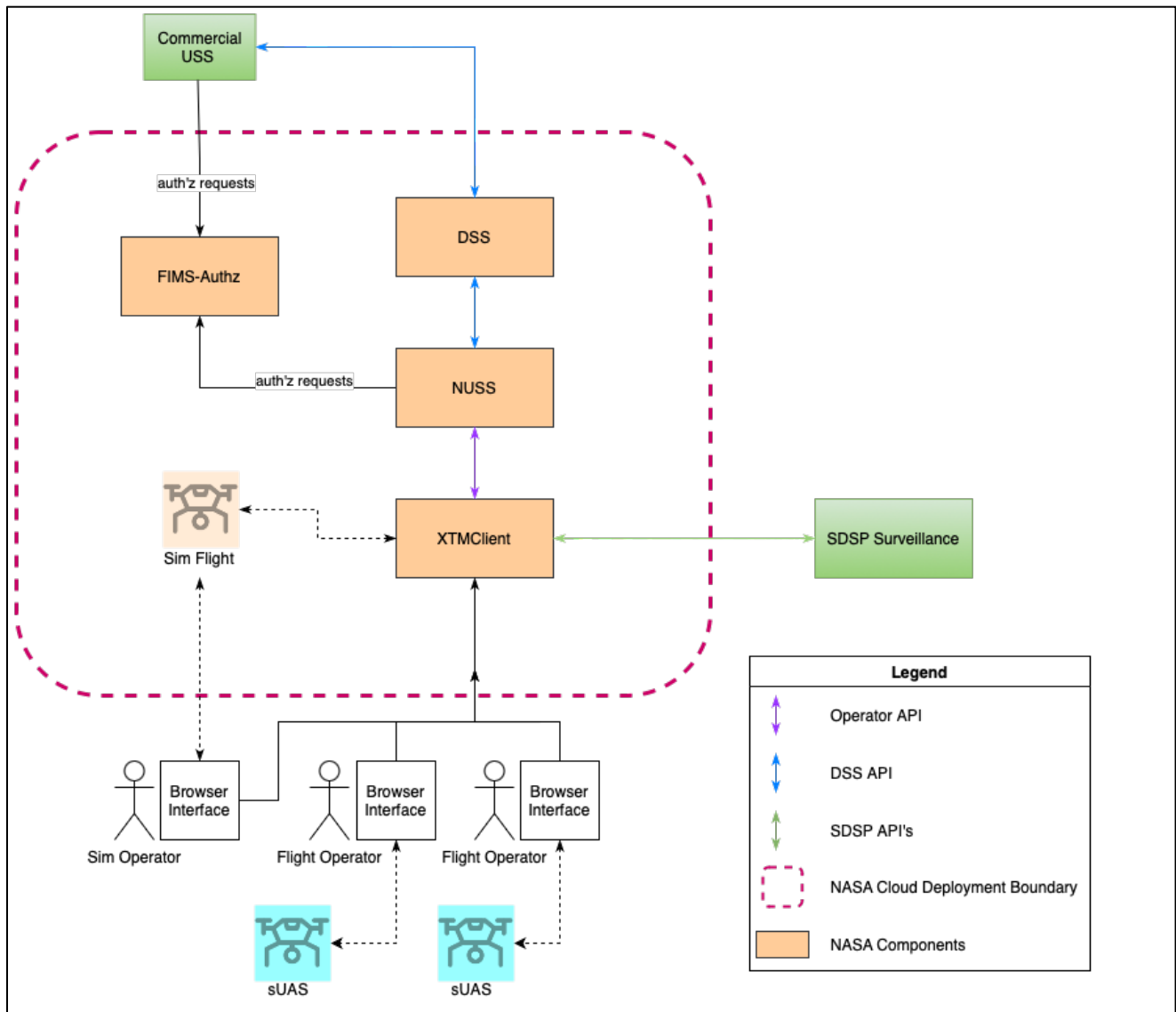


Figure B1. NASA USS functionality.

¹⁶ <https://github.com/interuss/dss>

1.2. Service Functions

1.2.1. Strategic Conflict Detection

Strategic Conflict Detection (SCD) is provided to our supported operators and is implemented per the UA-UA SDD¹⁷ defined by the US UTM Implementation Governance Agreement. SCD is used while planning a future flight, activating a flight, closing/cancelling flight. Details regarding this function are deferred to the governance agreement upon which this function is based.

1.2.2. Conformance Alerting

NASA USS's conformance alerting feature will determine an operation to be conforming when the vehicle's position report is inside the operation Intent's 4d geometry; and determine an operation to be conforming when the vehicle's position report is in time and is the same as a point on the edge an Operation's Intent 3-geometry. A flight can trigger non-conformance in other situations such as late position reports. Conformance limiting parameter are published in NASA's Application Programming Interfaces (API) Specification for Conformance Alerting.

Notifications related to conformance alerting are provided to the operator via the interfaces described in the "API Services" section below. These alerts are intended to provide the operator with real-time information about the conformance of the operation. The alerts are not shared with other USSs.

Conformance Monitoring and Alerting for Situational Awareness (CMSA) is a role in the "Standard Specification for UAS Traffic Management UAS Service Supplier" international standard ASTM F3548-21 (2022). CMSA to notify other USS of non-conformance is not standardized and, therefore, is not a service that is currently available.

1.2.3. Aggregate Conformance Monitoring

The Aggregate Conformance Monitoring (ACM) service collects metrics around an operator's interactions with CMSA to measure flight conformance on a per-operator basis. ACM gathers information on the conformance of all operational intents created by each operator (over a period of time or flight hours) to determine whether conformance requirements are met. ACM calculates statistics and summarizes an operator's performance, then sends a report to the user.

1.2.4. API Services

NASA USS provides Application Programming Interfaces (APIs) that allow for the use of various clients to interact with the various services. These APIs are the mechanism for implementations that meet the needs of the operator.

Distinct APIs provided by NASA USS are as follows:

1. Operator API: This REST interface provides the primary interactions with the NASA USS including the ability to submit new operation plans, modify operation plans, and submit position reports. Some stakeholders have used it to feed visual displays.
2. Dynamic Data Provider: A Secure Websockets streaming subscription interface where asynchronous messages are provided to the connected client. Its messages include information about operation non-conformance (used by Conformance Alerting) new operations from other USSs, and modifications to operations from other USSs.

¹⁷ https://github.com/utmimplementationus/getstarted/blob/main/docs/UA-UA_Strategic_Coordination_Service_Description_Document_v1.0.pdf

3. Subscription API: Allows the client to manage subscriptions within the DSS. Stakeholders can use this interface to provide situational awareness of operations from other USSs in its geographical area.
4. Authorization API: Allows the client to authenticate to NASA USS and receive authorization tokens which allow the interface to implement role-based access control.

1.2.5. XTMClient

NASA USS optionally provides a graphical user interface to its operators and other users to access the API services. This XTMClient leverages the Operator API to receive data from the operator, using graphical inputs. Visualization of operational data is provided in XTMClient, as well as indications of planning state and other data. Use of the XTMClient is not mandatory for an operator that may have developed its own user interface using NASA USS's API.

1.2.6. Telemetry Relay

Telemetry Relay is a program that works to conduct telemetry data from the vehicle directly to the USS (Freddie Services). The flow of information is as follows: The GCS forwards its telemetry information to the Telemetry Relay, the relay sends the information to the NASA USS, which then forwards it to the XTM client for display. In the chance that the relay application closes or stops working, telemetry will stop. The application can be restarted to resume telemetry. If browser is closed and/or reloaded, telemetry relay will continue to run and sending telemetry. XTM Client in browser can then be reconnected to telemetry relay without any interruption. The C2 of the UAS is not affected by the closing of the application.

1.2.7. User Notifications

NASA USS emits a user notification upon each operational state change. Each message is accompanied by explanatory text if needed. All Operator messages are defined in NASA USS's Operator Interface API specification.

In a nominal lifecycle of an operation, SCD sends the user notification of all state changes. These include nominal state changes such as those to and from Accepted, Activated, and Closed. SCD sends alerts for state changes such as failure of a request to Activate on operation.

Conformance Alerting will detect when an operation goes non-conforming and trigger a notification to the user as specified by the UTM Interface Spec (NUP Operator API version: 5.0.1).

Notification(s) are sent in the case of USS system failures such as loss of communication between user(s) and NASA USS, and loss of communication with other USSs and/or DSS (see Table B2). In such circumstances, the user notification may need to be sent over a channel outside of NASA USS, such as email or phone.

1.2.8. Authorization

NASA UTM allows a UTM authority to protect NASA UTM resources by verifying RFC 7519 role-based authorization data. Before it begins servicing a request, NASA USS will validate the Pilot in Command's token, which was obtained by the Freddie OAuth service. NASA USS will check token properties such as expiry.

1.2.9. Logging

NASA USS logs events in a comma-separated data format consistent across all log statements. Logs support ACM, operator archive needs, general system tracing, and other similar functions.

1.2.10. Entities

NASA provides end-to-end capability as the service provider. Its microservices architecture, shown in the Architecture section of this document, provides ease with scalability and functional flexibility.

1.2.11. Safety Benefit/Risk Mitigation

NPR7900 Chapter 5 Section 3 Aircraft Operations Management Procedural Requirements inclusive of Unmanned Aircraft Systems Operations is followed for risk assessment.

SCD is intended to provide the operator with a recognized mitigation for UA-UA hazards, per 8040.6A and the SDD.

Operator implements mitigations by cross-referencing airspace class and restrictions with operational plans. Operational intent is adjusted as needed to reach acceptable risk level.

Generally, the failure of NASA USS will result in a loss of efficiency, or a loss of confidence in the overall system but will not generally result in a loss of aircraft control. If NASA USS fails, operators may lose situational awareness of other aircraft flying in the same environment. However, each Pilot in Command will still have C2 of their UAS via ground control station and/or remote control. Active flights will have approved operations where volumes have been deconflicted and execution remains under pilot control.

1.2.12. Operational Risk Mitigation

Reference FAA Form 7711-1 2020-P107-WSA-18993-R for NASA Ames Research Center's COA from the FAA.

2. Area of Operations/Coverage

2.1. Geographic, Altitude, Coverage Limits

Since the NASA USS currently supports only internal operations, the area of operation is based on NASA facilities and test locations where NASA operators have obtained approvals. The altitude limits are 400 feet AGL or the limit for any approvals already obtained.

System limits on the coverage area of flight include several aspects. The maximum Operation Intent Volume area is limited to 2500 km² due to the limitations on the Discovery and Synchronization Service (DSS)⁴. NASA USS limits the total number of vertices across all volumes comprising an operational intent to 1000. NASA USS enforces an altitude minimum –8,000 meter and maximum 100,000 meter. WGS84 is the coordinate reference frame.

NASA USS has been tested at 400 ft and below at the following locations: Virginia Tech, Virginia; Rome, New York; Corpus Cristi, Texas; Downtown Reno, Nevada; Grand Forks, North Dakota; Fairbanks Alaska; NASA Ames Research Center, California.

⁴ <https://github.com/interuss/dss/blob/master/pkg/geo/s2.go#L22>

2.2. Hours of Operation

The NASA USS is available continuously to support operation planning when needed. Scheduled downtimes are announced to internal operators and are planned when disruption will be minimal.

3. Standards Initiated by Service

NASA USS is conforming to the following standards:

- ASTM-F3548-21 Strategic Coordination Compliance Matrix
- UA-UA Strategic Coordination Service Description Document (SDD)

4. Service Architecture and Information Flows

All NASA USS services use an internal messaging system implemented by a broker. The broker provides scalability and extensibility of Freddie's internal messages. The internal broker architecture provides asynchronous messaging. The broker implementation is opaque to users and other stakeholders.

Each microservice is independently deployable, and each service can be configured at deploy time.

Figure B2 illustrates the way that NASA USS provides the services listed in Section I. Operators communicate with NASA USS via the API Services described above. The NASA USS then communicates with other USSs through APIs standardized by ASTM. As for SDSP and the aircraft control platform, often the interfaces with NASA UTM are vendor specific and proprietary. Data flows are controlled within NASA based on System Security Plans (SSPs). In cases where data must flow between systems dictated by separate SSPs from an operator to the NASA USS across differing SSPs, an Interconnect Service Agreement (ISA) is required. This is common within NASA when systems are owned and managed by different teams or facilities.

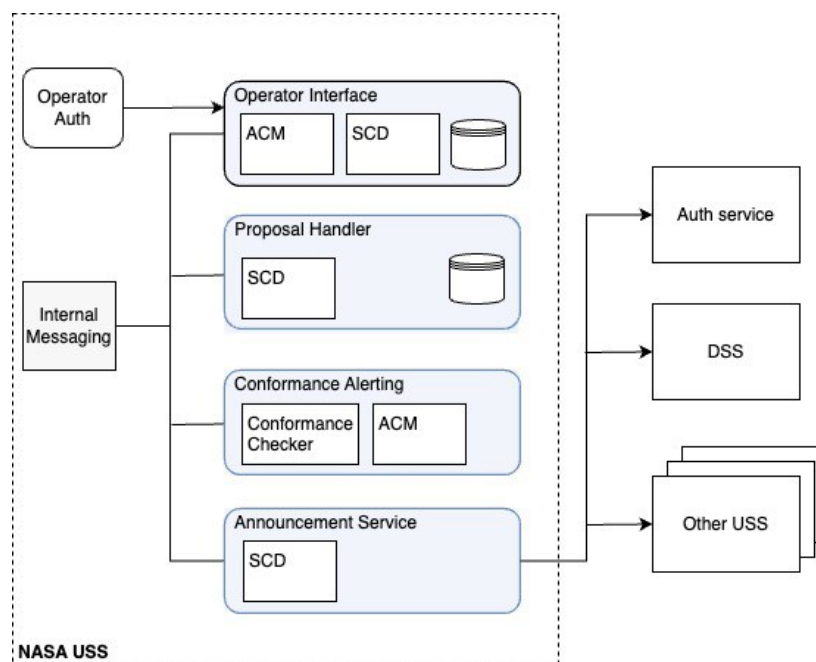


Figure B2. Functional block diagram - NASA USS (Blue)

The Operator Interface service maintains operational state and has its own database. When the Operator submits a plan, the plan goes to the Proposal Handler. Proposal Handler provides the SCD functionality where a Proposed operation is evaluated against the airspace before changing the operational state to Accepted. The Announcement Service allows the NASA USS to interface with other USS Service Providers via the InterUSS USS-USS and USS-DSS APIs.

An operation's telemetry is checked against its Operational Intent to provide conformance alerting, and this occurs within the Conformance Alerting service. If the operation is outside the operational intent, the operator is notified of the discrepancy via alert message.

NASA USS's SCD Service is implemented according to the international standard ASTM F3548.

4.1. External Hardware, Data Sources, or Other Information

NASA USS does not require external hardware for the operators to use the service. However, operators may bring additional tools to support their operations which will be outlined/addressed in the ConOps.

NASA USS does not incorporate special or secondary data sources to function.

4.2. Software and/or Hardware Management (including changes and updates)

Software development is planned and managed according to customer agreed statement of work, per decision by NASA Chief Engineer and Capability Lead, in compliance with NASA Procedural Requirements 7150.2C NASA Software Engineering Requirements which outlines the following NASA required procedures for software development.

1. Software Requirements
2. Software Architecture
3. Software Design
4. Software Implementation
5. Software Testing
6. Software Deployment and Maintenance
7. Software Assurance

These processes are documented in NASA Ames Research Center's Aviation Systems Division's Facility Software Development Management Plan (SDMP) and Software Assurance Plan (SAP).

Software revision and update is managed based on the facility Software Configuration Management Plan (SCMP) which is also based on NPR 7150.2C.

The SCMP summarizes the following required categories.

1. Software Repositories and Versioning
2. Baselineing
3. Change Configuration Board
4. Release Management

Software source code repositories are hosted on github.com as the NASA organization. Software is archived in version control via a git repository. Scripts and tools used to automatically generate source code are versioned and managed in the same way.

The Quality Management System for the NASA USS is guided by NPR 7150.2C and NASA Standard 8739.8B Software Assurance and Software Safety Standard. To adherence to these standards, software development and assurance processes have been developed to ensure the quality of the NASA USS. These processes included, but are not limited to:

- Software code coverage to ensure the breadth of software testing
- Software assurance compliance matrix and metrics reporting
- Software testing plan that includes unit testing, system testing, integration testing, load testing, user acceptance testing, and security testing
- Reviews to maintain stakeholder expectations and acceptance criteria
- Software bug reporting
- Risk management

5. Usage Limitations

The primary purpose of the NASA USS currently is to provide the Strategic Coordination service to NASA operators that require its use. All other flight and safety requirements are still incumbent upon the operator that is using the NASA USS for this purpose. Thus, the NASA USS does not place any other limitations on the operator. Expectations are outlined in the SLA.

As an example, this is a non-exhaustive list of checks that NASA USS does *not* perform for an operator:

1. Appropriate airspace class
2. UA limits such as weight, speed, endurance
3. Current restrictions in the airspace
4. Weather conditions

Elements such as these are for the operator to check for safe operation. The NASA USS will provide Strategic Coordination when the operator deems it is required.

5.1. Types of Operators

NASA is currently the sole operator supported by NASA USS. NASA-approved operators are under Part 107 and anticipated Part 108. Additionally, NASA operators must also complete internal Safety Review Board. NASA USS supports all types of civil operators.

5.2. Types of Aircraft

There are no limits to the types of vehicles that are supported by NASA USS. The service is agnostic to the specific vehicle type. See the associated ConOps for operator's vehicle specifics.

This ConUse applies to sUAS vehicle as defined by the FAA. The operator must be capable of providing the position of its operations via the provided APIs at 1 Hz while in the Active state.

6. Degradation and Failures

NASA USS Operational State changes meet the ASTM requirements. Table B1 describes the participants in an active UTM Service.

Table B1. Participants in the context of Degradation States	
<i>Participant Name</i>	<i>Role</i>
Pilot in Command	Person controlling the sUAS whose actions are tactical
UAS Operator (or Operator)	Person in charge of the flight/operational data, who determines its strategic schedule. Responsible for safe operations within all operational constraints.
UAS Services Supplier (USS)	Public or private service that participates in the USS Network.
NASA USS Administrator	Person who maintains the NASA USS products and its deployments
Material Assets	The vehicle, its ground controls station, batteries and other equipment managed by the Pilot in Command and the Operator.

Table B2 shows Degraded states, their impact on the participant and their mitigation or workaround until the problem is solved. Note that this analysis is for a single NASA USS system; with redundant NASA USS systems (such as is implemented with a green/blue strategy), the degraded state would be quite different.

Table B2. NASA USS Degradation SCD and their Safety Effects on Participants			
<i>"Given That" SCD is Processing this Request</i>	<i>Degraded Behavior: The Return Status of this API</i>	<i>Impact to Participant</i>	<i>Mitigation Until Problem is Understood and/or Solved</i>
Propose new operation	valid 5xx due to an internal error	Can't add new operation	none
Propose new operation	OperatorAPI slow response time due to slower service time in SCD	delayed adding new operation	reduce load (request rate)
Propose new operation	OperatorAPI timeout due to problem in the SCD service	Can't add new operation	none
Propose new operation	OperatorAPI slow response due to slower service time in the Operational Environment	delayed adding new operation	reduce load
Propose new operation	OperatorAPI timeout due to problem in the SCD service	Can't add new operation	none
Modify operation	valid 5xx due to an internal error	Can't modify an operation	if flying, land vehicle(s)
Modify operation	OperatorAPI slow response time due to problem in the SCD service	delayed OpMod	reduce load; if flying, possibly land vehicle(s)
Modify operation	OperatorAPI timeout due to problem the SCD service	Can't OpMod	if flying, land vehicle(s)
Modify operation	OperatorAPI slow response due to problem in Operational Environment	delayed OpMod	Operator can reduce load
Propose new operations or Modify operation at a high rate	slower response time in the SCD service	slower response time	reduce load

Errors described as 4xx or 5xx are referring to codes from the Internet Engineering Task Force (IETF) Request for Comments (RFCs), where a 4xx status code indicates that the client seems to have erred and 5xx indicates a service error. For an SCD error, the problem would be in the Operator API, Proposal Handler, or DSS services. The Operational Environment refers to the cloud hosting infrastructure, data storage and messaging services which support NASA USS.

7. Cybersecurity Considerations

7.1. Cyber Threat Analysis

NASA's initial development of the USS began with analysis of the threats to the overall UTM system. The threat modeling approach was a combination STRIDE and DREAD methods. Stakeholders identified threats to the system and the affected components. The STRIDE method categorizes the threats as Spoofing, Tampering, Information Disclosure, Denial of Service, or Escalation of Privileges. The stakeholder scores the threat using the DREAD approach. On a scale of 1–10 for the categories of

Damage, Reproducibility, Exploitability, Affected Users, and Discoverability. Given the categorization and the scoring, NASA identified the highest value concerns.

NASA developed the initial authorization and authentication requirements for UTM and published them in a technical memo¹⁸. Since the initial analysis, further stakeholder analysis has been performed with support of NASA. The report for that activity is published (UTM Pilot Program [UPP]¹⁹) as part of the UTM Pilot Program and became an influential document in the reasoning about cybersecurity and UTM.

All this input fed into the development of the ASTM F3548-21 standard and the industry approach to discovery, authentication, and authorization.

As a deployed system, all NASA software meets NASA cybersecurity requirements. These requirements are mostly derived from the security and privacy controls defined in NIST 800-53. The appropriate controls are determined based on the categorization of the system based on the sensitivity of the data which it handles.

The process for determining the categorization of NASA USS follows NIST FIPS 199. NASA has classified NASA USS as a “Moderate” system given the data that NASA USS handles. To achieve this rating, the primary information types were cataloged. Those types are user profile data, UAS operation data, map data, vehicle performance data, configuration files, and other third-party data. Each of these data categories were then assigned a security category based on Confidentiality, Integrity, and Availability (CIA). For each data type and each security category, the assignment of Moderate was established as a baseline. This is the approach to define NASA USS as a “Moderate” system overall.

In addition, NASA USS followed NIST SP 800-122 to determine the PII Confidentiality Impact level. Based on the data categories, the analysis led to a “Moderate” categorization for the PII Confidentiality Impact Level.

Various NASA IT requirements based on this classification protect NASA USS and its environment. These include, but are not limited to:

1. For Production system, use of virtual public clouds to control ingress and egress of data from and to appropriate systems.
2. Active monitoring of network behavior and system behavior. Monitor data integrity if required.
3. A 24/7 SOC and NOC. If available 24/7 on-site Operations.
4. Regular operating system patching, defined by the NASA Cybersecurity Standards and Engineering Team (CSET).
5. Amazon Web Services (AWS) best practices.
6. Credential management.

¹⁸ <https://ntrs.nasa.gov/api/citations/20190032004/downloads/20190032004.pdf>

¹⁹ https://www.faa.gov/sites/faa.gov/files/uas/research_development/traffic_management/utm_pilot_program/UPP1_Technical_Report_version2.0.pdf

At the software level, the team employs the following approaches:

1. Dependency scanning. All libraries and dependencies are scanned at the manifest level, with alerts and tasks created for the team when a library has a known vulnerability. The development team fixes vulnerabilities with a top-level priority.
2. Source Code Scanning²⁰. Insecure coding practices are highlighted by regular source code scanning, with alerts available to the team including recommendations for remediation.
3. Secure Coding Practices. NASA has developed in-house secure coding practices which are documented and embraced by the team. The storage of credentials and keys are designed to be outside of the code and secure.
4. Robust testing. Unit tests, system tests, and performance tests provide further robustness against cyber threats.
5. All code changes of any size require two peers to code review.

7.2. Worst Credible Outcome

Results of a Failure Mode and Effects Analysis (FMEA) were provided to the FAA.

7.3. Credentials

The NASA USS Administrator sequesters passwords in a separate repository and deploys them on an as-needed basis. A compromised credential could be obtained through a social engineering attack or by administrator error.

Mitigation of such attacks includes credential rotation, password management procedures and a minimal set of actors granted password management privileges. All NASA employees complete mandatory security training on a regular basis and are aware of attacks and their prevention.

7.4. Message and User Authentication

While NASA is an individual operator from the perspective of the external UTM network, internally there are different teams and Centers which function like separate operators. Consequently, NASA USS allows for the definition of an “Operator” that is allowed any number of authentication credentials. Each set of credentials for a system or a person will be associated with exactly one such operator. NASA USS follows a Role-Based Access Control (RBAC) approach so that each set of credentials comes with a set of well-defined roles and each role has a specific set of permissions.

²⁰ <https://docs.sonarsource.com/sonarqube/latest/>